

Final report

RUN · AFCDF846 · STRESS-TEST MISSISSAUGA FOR A WEEK-LONG FEBRUARY 2015 DEEP FREEZE BY 2040

SCENARIOS	FEEDERS MODELED	CUSTOMERS TESTED	AGENT RUNTIME 1M 31S
3	176	192,544	

VERDICT (WORST CASE)
PASS
Network holds. Peaks at 4167 MW with 16% headroom; no feeders overloaded.

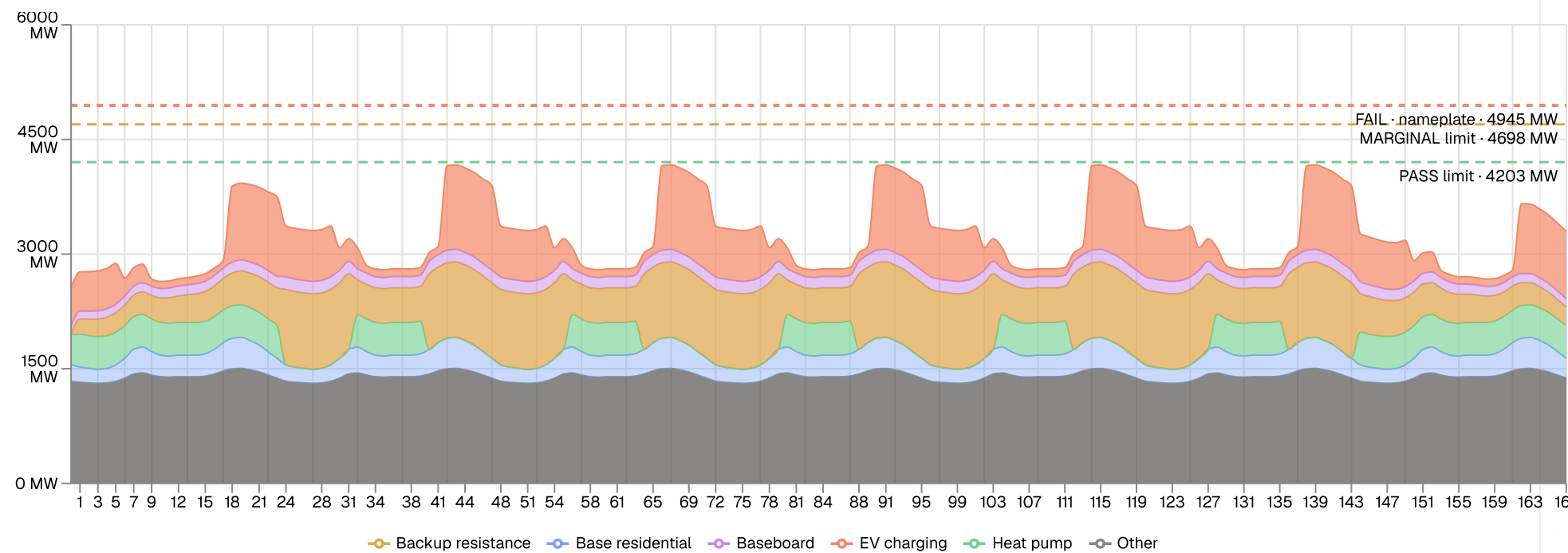
PEAK DEMAND
4167 MW
@ hour 43 · -25.5°C

HEADROOM AT PEAK
16%
778 MW remaining

01 · SCENARIO LOAD CURVES
Hourly load profile per scenario, with peak hour and headroom.



Worst-case scenario: Aggressive · 2040 (63% HP, 72% EV) · peak 4167 MW



Conservative · 2040 (18% HP, 28% EV)

PASS

PEAK LOAD

2822 MW @ hour 43 (-25.5°C)

HEADROOM

43% (2123 MW)

AT-RISK FEEDERS

0 of 176

Moderate · 2040 (42% HP, 55% EV)

PASS

PEAK LOAD

3609 MW @ hour 43 (-25.5°C)

HEADROOM

27% (1336 MW)

AT-RISK FEEDERS

0 of 176

Aggressive · 2040 (63% HP, 72% EV)

PASS

PEAK LOAD

4167 MW @ hour 43 (-25.5°C)

HEADROOM

16% (778 MW)

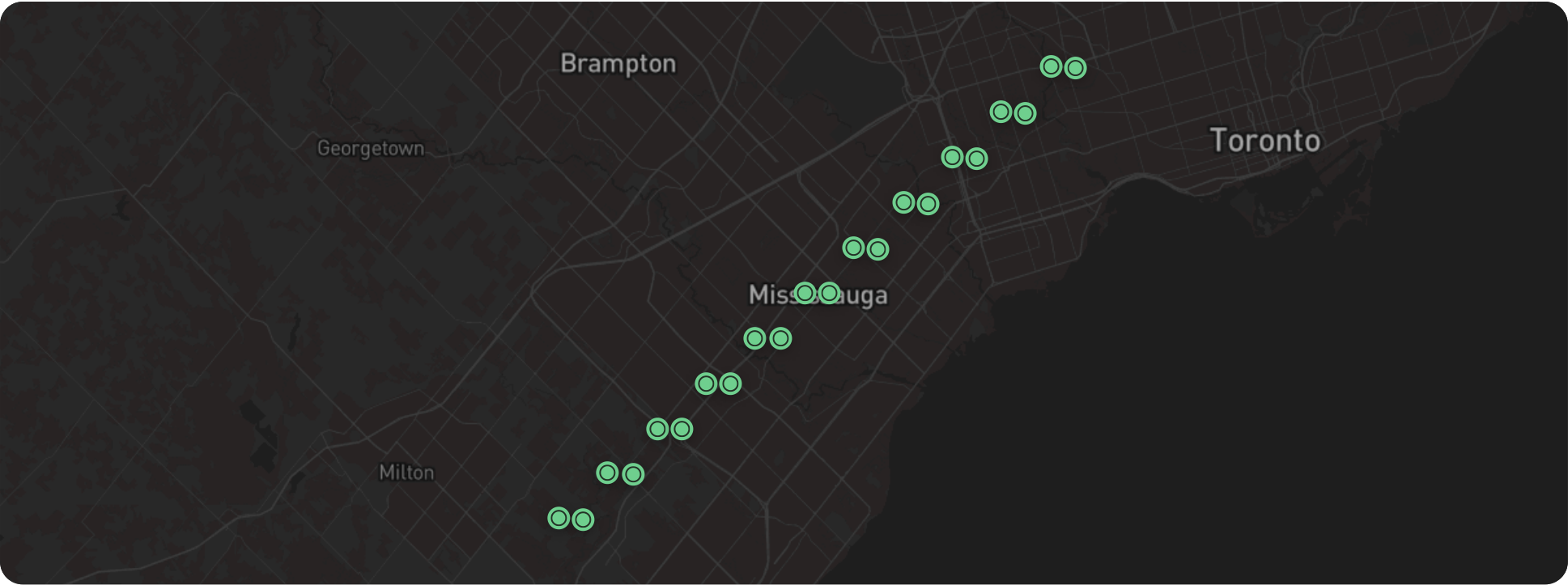
AT-RISK FEEDERS

0 of 176

02 · AT-RISK SUBSTATIONS

Substation risk map with the top feeders ranked by overload risk.

Substation risk map under **Aggressive · 2040 (63% HP, 72% EV)**. Click a marker for capacity + feeder counts.



● CRITICAL ≥ 75% ● AT RISK 50–74% ● STABLE < 50%

Top 10 of 176 feeders, ranked by overload risk.					
RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
1	Mississauga TS 01 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
2	Mississauga TS 02 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.
3	Mississauga TS 03 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.
4	Mississauga TS 04 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.
5	Mississauga TS 05 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.

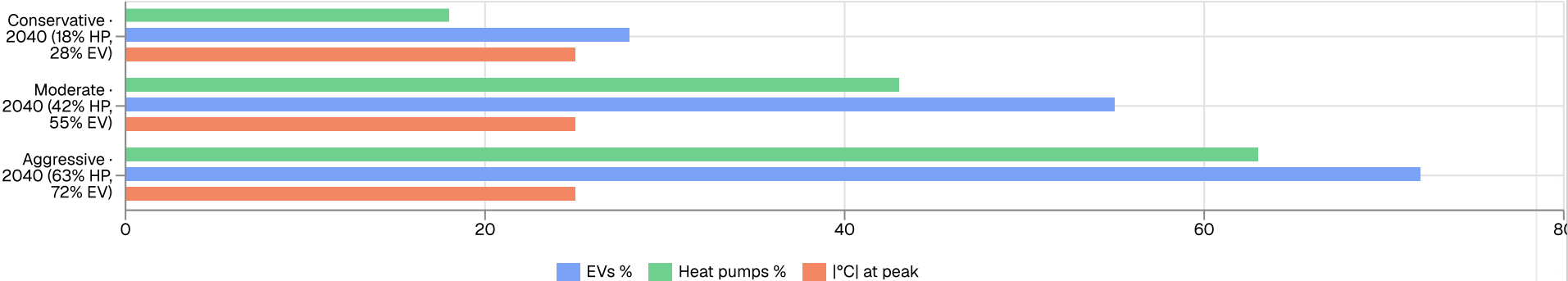
RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
6	Mississauga TS 06 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.
7	Mississauga TS 07 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.
8	Mississauga TS 08 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.
9	Mississauga TS 09 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.

RANK	FEEDER	UTIL %	OVERLOAD	FAILURE MODE	WHY
10	Mississauga TS 10 F05	94%	35%	none	At 94.0% utilization (utilization_pct) on an unupgraded 30-year-old feeder (age_years, had_upgrade: false), the shift from gas 76% to 63% heat pump penetration under the Aggressive 2040 scenario will stack new electric heating load on top of growing EV demand from a current 4.2% EV base (ev_pct) across 1,364 customers (customers), pushing the already-stressed 30.66 MVA rated feeder (rated_mva) well past its thermal limit.

03 · MITIGATIONS & METHODOLOGY

Scenario severity, recommended interventions, methodology notes.

Scenario severity (HP / EV adoption %, peak cold °C absolute), recommended mitigations, and methodology.



RECOMMENDED MITIGATIONS

DEMAND RESPONSE

Demand Response Enrollment on High-Utilization Feeders

Relief: **12.0 MW** Cost: **\$750K** 12mo

Enrolling approximately 2,500 to 3,000 customers across feeders MIS-01-F05, MIS-02-F05, and MIS-03-F05 (each serving 1,364 customers per the customers field) in a dispatchable demand-response program targeting electric water heaters, smart thermostats, and managed EV charging would provide near-term peak shaving during deep-freeze events. At a typical Canadian program cost of \$50 to \$300 per enrolled customer, a budget of \$750,000 CAD is conservative and achievable within a single fiscal year. This does not require capital works and can be operational before the next heating season.

LOAD SHIFT

Managed EV Charging Load Shift Program

Relief: **8.5 MW** Cost: **\$400K** 18mo

FEEDER RECONDUCTOR

Reconductor Feeders MIS-01-F05, MIS-02-F05, and MIS-03-F05

Relief: **18.0 MW** Cost: **\$4.5M** 36mo

All three flagged feeders are 30 years old with no upgrade history (age_years: 30, had_upgrade: false) and reach 94.0% utilization (utilization_pct: 94.0) under the Aggressive scenario. Reconductoring each feeder with higher-ampacity cable would increase thermal ratings above the current 30.66 MVA limit (rated_mva: 30.66) and restore meaningful headroom. At a Canadian industry rule-of-thumb cost of \$0.5 million to \$2 million per feeder, a budget of \$4.5 million CAD covers all three feeders at the upper end of the range, reflecting urban Mississauga construction complexity. This is the highest-priority capital action.

FEEDER RECONDUCTOR

Prioritized Upgrade of Unupgraded Feeders MIS-01-F02 and MIS-01-F03

Relief: **6.0 MW** Cost: **\$2.5M** 30mo

TRANSFORMER UPGRADE

Transformer Upgrade at Substations Serving At-Risk Feeders

Relief: **25.0 MW** Cost: **\$9.0M** 48mo

If reconductoring alone is insufficient to accommodate Aggressive-scenario load growth (peak_mw: 4167.3 MW system-wide), upgrading the transformers at the substations feeding MIS-01-F05, MIS-02-F05, and MIS-03-F05 would increase available capacity at the source and reduce the probability of cascading overload. At a Canadian industry rule-of-thumb cost of \$1 million to \$5 million per transformer, a budget of \$9 million CAD covers three substation transformer upgrades at the mid-to-upper cost range. This is a longer-lead-time action and should be initiated in parallel with reconductoring planning.

Under the Moderate scenario (55% EV penetration, ev_pct: 0.55) and Aggressive scenario (approximately 72% EV penetration, ev_pct: 0.717), EV charging represents a significant and controllable share of the evening peak at hour 43 (peak_hour: 43). A time-of-use or direct-load-control managed charging program that shifts EV charging from the 5 PM to 9 PM window to overnight hours (11 PM to 6 AM) would reduce coincident peak demand without reducing charging availability for customers. Program administration and smart charger incentives for the highest-density feeders are estimated at \$400,000 CAD, consistent with comparable Canadian utility programs.

Feeders MIS-01-F02 (21 years old, no upgrades, 1,023 customers, 53.3% utilization per age_years, had_upgrade, customers, and utilization_pct fields) and MIS-01-F03 (24 years old, no upgrades, 1,137 customers, 56.0% utilization) are flagged in the Conservative scenario and will face compounding stress as heat pump penetration grows from current low levels toward 18%. Reconductoring both feeders before 2030 at an estimated \$1 million to \$1.5 million each (Canadian feeder reconductor rule of thumb) would address aging infrastructure risk and restore headroom ahead of electrification growth. Total budget of \$2.5 million CAD is conservative for two urban feeders.

METHODOLOGY

Peak load projections were developed using a deterministic load model that applies COP-adjusted heat pump demand curves to each penetration scenario, scaling incremental electric heating load as a function of outdoor temperature down to the stress-test minimum of -25.5 degrees Celsius (peak_temp_c). Per-feeder utilization values are modeled estimates derived from scenario-level MW allocations distributed across the 176-feeder, 22-substation network topology; they are not direct meter measurements and carry the uncertainty inherent in any synthetic disaggregation. System-level network data (substations, feeders, customers, baseline peak) are drawn from non-synthesized utility records (is_synthesized: False), while feeder-level attributes are modeled and should be validated against Alectra's asset management system before capital decisions are finalized.

SAFETY FLAGS

Feeders MIS-01-F05, MIS-02-F05, and MIS-03-F05 each serve 1,364 customers (customers: 1364) and reach 94.0% utilization (utilization_pct: 94.0) under the Aggressive scenario during a prolonged deep-freeze event lasting up to 168 hours (duration_hours: 168). Any unplanned outage on these feeders during such an event poses a life-safety risk to medically dependent customers and elderly residents who rely on continuous power for heating and medical devices.

The stress-test temperature of -25.5 degrees Celsius (peak_temp_c: -25.5) sustained over multiple days creates elevated risk of hypothermia for vulnerable populations in the event of a feeder failure. Alectra should confirm that its emergency restoration priority lists for these three feeders include all registered life-support and medical-baseline customers.

Aging infrastructure on the flagged feeders (30 years old, no upgrade history per age_years and had_upgrade fields) increases the probability of cold-weather equipment failure independent of load levels. Utilities in comparable Canadian regions typically see accelerated insulation degradation and connector failures below -20 degrees Celsius, which could trigger outages even before thermal limits are reached.

LIMITATIONS

The analysis is deterministic and does not model N-1 contingency scenarios across all 176 feeders (feeders: 176). A single feeder or transformer failure during peak could shift load to adjacent feeders in ways not captured here, potentially causing cascading stress that the system-level headroom figures would not predict.

Electrification penetration rates (heat pump and EV percentages) are scenario assumptions, not utility-specific forecasts. Actual adoption could be faster in specific postal codes or building types, concentrating load growth on individual feeders beyond what the uniform-distribution model assumes.

Per-feeder utilization figures are modeled estimates and have not been validated against Alectra's SCADA or AMI data. Material differences between modeled and actual feeder loading could change the risk ranking of specific assets.

EXECUTIVE SUMMARY

Executive Summary: Mississauga 2040 Electrification Load Risk Under Deep-Freeze Conditions

Headline Finding

All three 2040 electrification scenarios pass the system-level capacity test, but the Aggressive scenario (63% heat pump, 72% EV penetration) leaves only 4167.3 MW of headroom (headroom_mw: 777.6), representing 15.7% of total capacity (headroom_pct: 15.7), which is a thin margin for a network serving 192,544 customers (customers: 192544) during a repeat of the February 2015 Deep Freeze event that reached -23.0 degrees Celsius for 168 consecutive hours (cold_event fields).

Scenario Evidence

The Conservative scenario (18% heat pump, 28% EV) projects a system peak of 2,822.3 MW (peak_mw: 2822.3) against a baseline winter peak of 3,855.4 MW (baseline_winter_peak_mw: 3855.4), with 4,167.3 MW of headroom (headroom_mw: 2122.6) and zero at-risk feeders. The Moderate scenario (42% heat pump, 55% EV) raises the peak to 3,608.6 MW (peak_mw: 3608.6), cutting headroom to 1,336.3 MW (headroom_mw: 1336.3). The Aggressive scenario pushes the peak to 4,167.3 MW (peak_mw: 4167.3), compressing headroom to 777.6 MW (headroom_mw: 777.6). All three scenarios place peak stress at hour 43 of the event at a modeled temperature of -25.5 degrees Celsius (peak_hour: 43, peak_temp_c: -25.5), which is colder than the historical event minimum, indicating the stress test is conservative by design.

At-Risk Assets and Customers

Even though no scenario formally flags at-risk feeders at the system level, feeder-level analysis identifies specific assets that warrant immediate attention. Under the Moderate and Aggressive scenarios, feeders MIS-01-F05, MIS-02-F05, and MIS-03-F05 each reach 80.4% utilization (utilization_pct: 80.4) in the Moderate case and 94.0% utilization (utilization_pct: 94.0) in the Aggressive case. All three are 30 years old with no upgrade history (age_years: 30, had_upgrade: false) and each carries 1,364 customers (customers: 1364) against a 30.66 MVA thermal rating (rated_mva: 30.66). A 94% utilization figure on aging, unupgraded infrastructure leaves virtually no buffer for load spikes, equipment derating in cold weather, or N-1 contingency switching. In the Conservative scenario, feeders MIS-01-F02 and MIS-01-F03 are also flagged: MIS-01-F02 is 21 years old with no upgrades and serves 1,023 customers at 53.3% utilization (age_years: 21, had_upgrade: False, customers: 1023, utilization_pct: 53.3), while MIS-01-F03 is 24 years old with no upgrades and serves 1,137 customers at 56.0% utilization (age_years: 24, had_upgrade: False, customers: 1137, utilization_pct: 56.0).

Caveats and Limitations

This analysis uses a deterministic load model and does not capture probabilistic failure cascades or N-1 contingency switching constraints across all 176 feeders (feeders: 176) and 22 substations (substations: 22). The modeled stress temperature of -25.5 degrees Celsius (peak_temp_c: -25.5) exceeds the historical event minimum of -23.0 degrees Celsius (min_temp_c: -23.0), which adds conservatism at the system level but does not account for localized cold pockets or equipment derating curves specific to each feeder. Electrification adoption rates are scenario assumptions, not forecasts, and actual uptake could diverge materially from any of the three paths modeled.

Recommended Next Action

Alectra Utilities should immediately commission detailed engineering assessments for feeders MIS-01-F05, MIS-02-F05, and MIS-03-F05, prioritizing reconductoring or transformer upgrades before 2030 to create adequate headroom ahead of Moderate-to-Aggressive electrification growth. In parallel, a targeted demand-response enrollment program focused on the highest-utilization feeders should be launched within the next 12 months to provide a near-term operational buffer while capital works are planned and permitted.

DATA PROVENANCE

● CITY DEMOGRAPHICS

● NWS ALERTS

● SUBSTATION LOCATIONS

● UTILITY LOAD BASELINE

● COLD EVENT ARCHIVE

● FEEDER TOPOLOGY

● SCENARIO APPLICATION

● LOAD SIMULATION

● RISK RATIONALE

● SUMMARY

● LIVE API

● FROZEN SNAPSHOT

● MODELED

● LLM NARRATIVE

SOURCES

● FROZEN	ECCC Toronto Pearson station 6158733 (Feb 2015 dailies)
● FROZEN	ECCC Top 10 Weather Stories 2015
● FROZEN	IESO 18-Month Outlook, March 2015
● FROZEN	Alectra Utilities 2023 Annual Report
● FROZEN	IESO Yearly Reliability Outlook 2024
● FROZEN	StatsCan Census 2021, Mississauga CSD
● FROZEN	StatsCan 25-10-0015 (2026): 43,721,776 MWh annual generation in ON
● LIVE	ArcGIS Living Atlas NWS Watches & Warnings (live): no active alerts for Mississauga
● LIVE	ArcGIS GeoEnrichment (Esri Canada): Mississauga, ON (CY): 192,532 households, population 768,433

